

Automated Alignment Between Channel and Filter Cascade

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Abstract: We propose a new and cost-efficient method based on receiver-side spectrum monitoring to mitigate filter impairments. We experimentally demonstrated up to 3dB-reduction in filter penalties for 32GBd PDM-QPSK signals.

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1. Introduction

The industry is promoting transponders with high symbol rates such as 43GBd and 64GBd to meet increasing channel capacity needs. This evolution provides a unique opportunity to optimize optical bandwidth usage - and thus costs - by closely matching channel symbol rate with allocated slot-size. The only limitations to this spectrum densification are the impairments from tight filtering at the reconfigurable optical add-drop multiplexers (ROADMs). In this context, network studies [1] showed that the extra design margins required to reach sufficient signal quality despite filter effects significantly hinders the capacity gains otherwise achievable by reducing slot-size.

To counter these limitations, various schemes were proposed to mitigate filtering impairments, e.g. through advanced equalization [2] or spectral reshaping [3]. In this paper, we propose a radically new and cost-efficient approach to reduce filter penalties. It consists in automatically cancelling the dominant contribution to filter penalties and related design margins [4], i.e. the detuning between the transmitted signal and the filter cascade due to various biases from manufacturing, ageing and/or temperature sensitivity [5].

We start by generically describing the alignment method based on the monitoring of the signal-filter detuning inside the coherent receiver, and its potential deployment in current optical networks. Through experiments, we then assess the validity and performance gains of the method and illustrate its operation.

2. Method for the automated alignment of the channel with the filter cascade

The solution we propose fundamentally relies on a closed-loop control of the signal-filter detuning as measured by the coherent receiver (Rx). In a nutshell, those measurements are pushed to a controller ordering corrections of transmitter (Tx) frequency to maintain a detuning close to zero, thus permanently minimizing filter penalties.

For the control loop to operate correctly, the first requirement is a robust and accurate method to monitor the signal-filter detuning. Solutions based on optical spectral analyzers (OSAs) hypothetically deployed in routing nodes were proposed [6,7]. Though potentially interesting for the detection of filter-related soft-failures, their application to the automatic correction of signal-filter detuning could be impractical due to the excessive deployment cost of the low-noise OSAs required to that end. As a cost-efficient alternative to OSAs, we thus propose a method achieving unbiased monitoring of the signal-filter detuning inside the coherent receiver (Rx).

Since filter effects are mitigated by the equalizer, we recover the signal-filter detuning D_{SF} from the signal power spectral density PSD_{ADC} at the output of the analog-to-digital (ADC) block. Amongst the many metrics deriving from PSD_{ADC} providing an image of D_{SF} , e.g. centers at $-X$ dB [7], we found the center of mass M_1 defined below to have the best metrological properties in terms of linearity, noise sensitivity and specificity to the signal-filter detuning:

$$M_1 = \int_{-F_s/2}^{F_s/2} f \cdot PSD_{PostADC}(f) \cdot df / \int_{-F_s/2}^{F_s/2} PSD_{PostADC}(f) \cdot df \quad (1)$$

where F_s is the ADC's sampling frequency and f is the baseband frequency. As ideal for the control loop, we should always verify $M_1=0$ when the transmitted signal is perfectly aligned with the filter cascade, i.e. when $D_{SF}=0$. However, because of the combined action of the coherent mixer and the opto-electronic frontend, a mismatch between Tx frequency F_{Tx} and the local oscillator frequency F_{Rx} , leads to $M_1 \neq 0$ even without signal-filter detuning. Optimal signal-filter detuning monitoring and consequently control loop operation is therefore obtained when $F_{Rx}=F_{Tx}$ is actively maintained. The next control loop requirement is for the controller to minimize M_1 thus D_{SF} by applying Tx frequency corrections. As a 1D convex optimization problem, convergence towards optimal signal-filter alignment through the minimization of $|M_1|$ can be efficiently achieved by many well-known algorithms, whose comparative performances are out of the scope of this paper. Maximizing a performance metric such as signal-to-noise ratio (SNR) instead of minimizing M_1 could appear as a more direct technique to achieve comparable signal-filter alignment. However, performance variations unrelated to frequency corrections would be likely to occur during the search for optimal correction, disturbing its convergence. In contrast, we designed the metric M_1 to only depend on the signal-filter detuning D_{SF} , making M_1 -based convergence robust to any kind of external perturbations. Additionally, the sign of

M_1 directly indicates the search direction, e.g. $M_1 > 0$ means that F_{TX} and F_{RX} must be increased to cancel the detuning. For lightpaths with performances close to the FEC limit, this property prevents any critical performance variation during the search. Finally, since M_1 monitoring does not depend on subsequent stages of the digital signal processing (DSP), this functionality is not affected by potential DSP failures. This is especially useful to diagnose and recover from an outage due to a failure of the clock recovery where the underlying cause is an excessive filter-laser detuning.

We further extend the control algorithm with conditional events to guarantee smooth network operation. To execute frequency corrections only when necessary, we trigger optimal searches only when $|M_1| > \text{ToI}_{M_1}$, ToI_{M_1} defining a critical signal-filter alignment. Stop conditions are also required to cancel the potential risk of channel collisions during or after the search for signal-filter alignment. To that aim, we strictly enforce guard-band-dependent limitations of the maximal frequency correction. With such control structure, we experimentally demonstrated the long-term SNR stability of a transmission despite major external perturbations of the signal-filter detuning (not represented).

In addition to a control plane architecture allowing closed control loop between paired Tx and Rx [6], a major requirement here is hitless fine-tunability of transponder frequencies F_{TX} and F_{RX} . On a hardware level, the external cavity lasers (ECLs) commonly integrated in commercial coherent transponders typically achieve hitless tunability on a 12GHz-range around their center frequency. Most current and future transponders should therefore handle hitless correction of the maximum signal-filter detuning expectable, i.e. $D_{SF, \max} = 4\text{GHz}$ according to standard specifications.

3. Experimental validation of the automated signal-filter alignment method

We perform a PDM-QPSK transmission at 32Gbd and RRC0.1 pulse shape with the experimental testbed depicted in Fig. 1a and further described in [8]. Both Tx and Rx lasers are ECLs and can be remotely tuned. We use a tunable optical filter (TOF) set on $F_0 = 193150\text{GHz}$ to emulate various amounts of cascaded filters through its bandwidth setting. We also study the impact of noise at reception with a variable optical attenuator (VOA) before amplification.

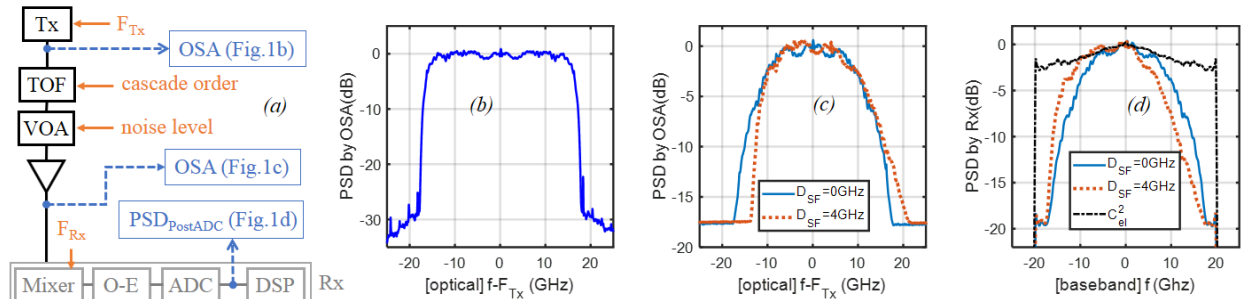


Fig. 1: (a) Experimental setup, spectrums (b) at Tx output, (b) just-before Rx, and (d) as monitored from Rx

The signal PSD measured at Tx output by an OSA (cf. Fig. 1b) is centered on Tx laser frequency F_{TX} and only depends on signal symbol rate and pulse shape. In contrast, the PSD measured just before Rx is deformed due to amplifier noise and filtering from the TOF. When we impose 4GHz of signal-filter detuning, the same PSD appears tilted (cf. Fig. 1c). Before analog-to-digital conversion (ADC), the signal is finally filtered by the opto-electronic frontend (cf. Fig. 1a), characterized by the filter function C_{el} (cf. Fig. 1d). To experimentally obtain PSD_{ADC} then M_1 through Eq. 1, we collect 2^{14} complex samples for each polarization at ADC output and use Welch's method with rectangular window and 0% overlapping to calculate the corresponding PSDs. These PSDs are finally averaged to obtain 2^{12} points of $\text{PSD}_{\text{PostADC}}$ (cf. Fig. 1d) corresponding to a baseband frequency resolution of 10MHz. The high ratio between the numbers of collected samples and PSD points ensures low noise PSD monitoring.

In our first experiment, we control the signal-filter detuning by setting the filter center at F_0 and shifting both Tx and Rx lasers by a varying frequency offset D_{SF} so that $F_{TX} = F_{RX} = F_0 + D_{SF}$. For each D_{SF} , we monitor corresponding values of M_1 and signal-to-noise ratio (SNR). We performed such measurements in two configurations: a 0dB attenuation with various filter bandwidth settings, and a 26GHz-wide filter with varying attenuation settings.

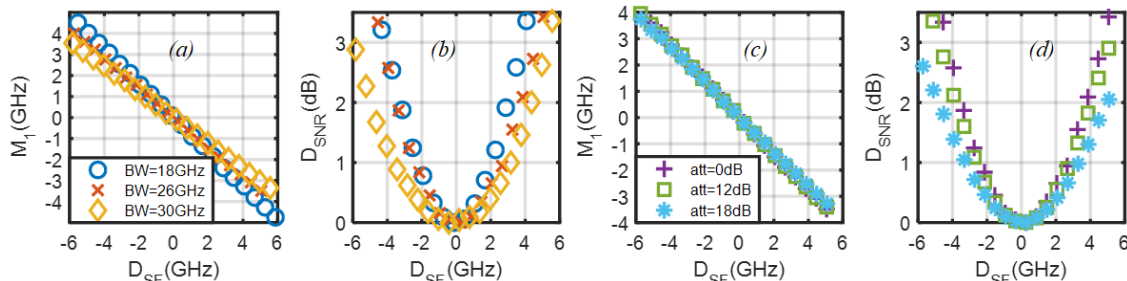


Fig. 2: M_1 vs signal-filter detuning D_{SF} and D_{SNR} vs $|M_1|$ for (a-b) VOA attenuation at 0dB and (c-d) TOF bandwidth at 26GHz

In Fig.2a&c, we plot M_1 as a function of the signal-filter detuning D_{SF} . We first observe the clear proportionality between M_1 and D_{SF} , with a slope S_{M1} depending on both filter bandwidth (Fig.2a) and attenuation level (Fig.2c). Note that $M_1(D_{SF}) \neq -M_1(-D_{SF})$ due to filter function asymmetry. We then plot $D_{SNR} = SNR_0 - SNR$ as a function of D_{SF} (cf. Fig.2b&d), where SNR_0 is the maximal SNR value reached for a given attenuation and filter bandwidth. The variability of the slope S_{M1} suggests that in actual networks, calibrations would be necessary to accurately quantify D_{SF} based on M_1 measurements. Regardless, we experimentally verify in Fig.2b&d that in all cases, the SNR is maximized when $M_1 \sim 0$, which makes the proposed monitoring technique optimal for automated signal-filter alignment.

Next, we experimentally validate the full solution using a basic convergence method where F_{Tx} and F_{Rx} are moved step-by-step, first to scan a wide range of M_1 values, then to reach the optimal correction as determined through the interpolation of $M_1 = g(F_{Tx} = F_{Rx})$. Modifying F_{Tx} and F_{Rx} by constant steps ΔF is experimentally convenient for setting the temporization between two frequency steps for lasers to stabilize. A typical search is illustrated in Fig.3b with an initial situation where $F_{Tx} \neq F_{Rx}$ and the signal-filter detuning has an arbitrary value. By leveraging monitoring of $D_{CFE} = F_{Tx} - F_{Rx}$ as provided by the carrier frequency estimation (CFE) block of the DSP, the algorithm first tunes the Rx laser, i.e. the local oscillator, to achieve $F_{Rx,1} \approx F_{Tx,1}$ and records $M_{1A} \approx -2.5$ GHz. Then, both Rx and Tx lasers are moved to $F_{Tx,2} \approx F_{Rx,2} \approx F_{Tx,1} + \mu M_{1A}$ where μ is set to 2.5. Based on recorded M_1 values, the algorithm finally stops at $F_{Tx,3} \approx F_{Rx,3} \approx F_{Tx,1} + 3.9$ GHz with an optimal SNR of 13.9 dB. PSDs before/after alignment are displayed in Fig.3a. We performed repeated tests with various initial signal-filter detunings D_{SF} and plot the resulting gains in SNR in Fig.3c.

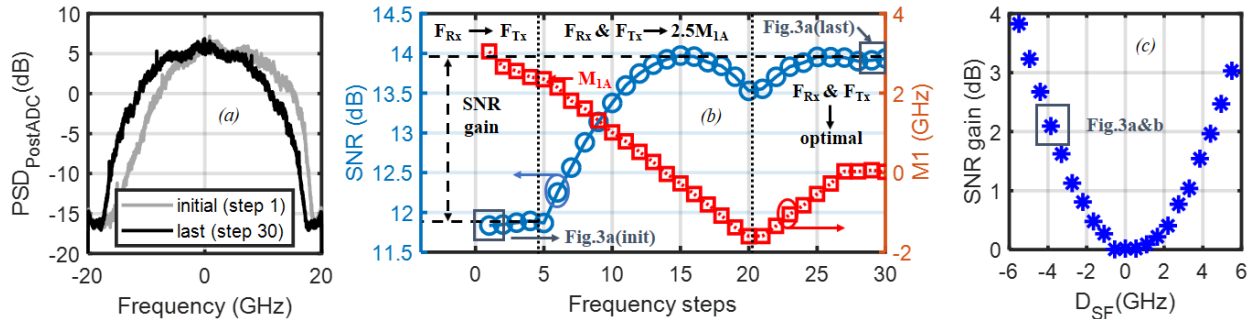


Fig.3: (a) Initial and last Rx-monitored PSD and (b) optimal frequency search with initial detunings of -3.85GHz for Tx and -4.35GHz for Rx (c) SNR gain observed after convergence with various initial Tx detunings D_{SF} ($F_{Tx,0} - F_{Rx,0} = 0.5$ GHz)

As expected, the SNR gains observed when correcting D_{SF} are comparable to D_{SNR} measurements in Fig.2b&d. Given that for typical network devices, F_{Tx} is known at ± 1.5 GHz and filter slot center is known at ± 2.5 GHz, signal-filter detunings up to $D_{SF,max} = 4$ GHz can realistically occur. For such detuning, results in Fig.2b&d show potential SNR gains up to 3 dB for BW=18 GHz, the equivalent of 5 cascaded route & select nodes with 37.5 GHz slot size.

5. Conclusion

We presented a method to monitor the detuning between transmitted signal and filter cascade inside the coherent receiver, and its application to the automatic cancellation of related filter impairments. We demonstrated that, for a 32 GBd PDM-QPSK transmission crossing an equivalent of up to 5 route & select nodes with 37.5 GHz slot size, our easily-deployable and cost-efficient solution, through its ability to counter naturally occurring signal-filter detunings of up to 4 GHz, can save up to 3 dB of filter margins as early as at the time of initial network design.

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6. References

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